AI/ML-Based Predictive Model for Ablation Zone Estimation in Tumor Treatment

Honours Project Report

March 2026

1. Introduction

1.1 Background

Cancer remains one of the leading causes of morbidity globally. Minimally invasive thermal ablation techniques - particularly Microwave Ablation (MWA) - have emerged as promising alternatives to traditional surgery for the treatment of solid tumors in organs such as the liver, lung, kidney, and adrenal glands. MWA works by delivering electromagnetic energy through interstitial antennas to heat and destroy cancerous tissue, creating a region of necrosis known as the ablation zone.

The shape and dimensions of this ablation zone are critically important: the zone must be large enough to completely cover the tumor while minimizing damage to surrounding healthy tissue. Currently, clinicians rely on a combination of clinical experience, manufacturer-provided ablation charts, and computational simulations (such as Finite Element Method models) to estimate the ablation zone prior to treatment. These approaches are either time-consuming, require specialized software, or fail to generalize across different antenna designs and operating conditions.

1.2 Problem Statement

> *Develop an AI/ML-based predictive model that accurately estimates the ablation zone produced during tumor treatment using combined hyperthermia and irreversible electroporation parameters, enabling faster and more accurate treatment planning while minimizing damage to surrounding healthy tissues.*

The central challenge is predicting the dimensions of the ablation zone (diameter, length, volume) from the treatment parameters (input power, duration, antenna type) without requiring computationally expensive physics-based simulations.

1.3 Objectives

1. Collect and curate ablation zone data from published experimental and simulation studies
2. Perform comprehensive exploratory data analysis to understand feature relationships
3. Engineer meaningful features from raw treatment parameters
4. Train and evaluate multiple ML regression models for ablation zone prediction
5. Compare model performance and identify the best-performing approach
6. Build an interactive prediction system for clinical decision support

1.4 Scope of Work

This project focuses on microwave ablation data collected from 30+ published research papers spanning 2004-2025. The dataset includes both experimental (ex vivo/in vivo) and simulation (FEM/computational) data across 15+ antenna types and a variety of power levels and treatment durations.

2. Literature Review

2.1 Microwave Ablation in Tumor Treatment

Microwave ablation employs electromagnetic waves (typically at 915 MHz or 2.45 GHz) to induce dielectric heating in biological tissue. Key treatment parameters include:

- Input Power (W): Energy delivered to the tissue via the antenna
- Treatment Duration (minutes): Time of energy application
- Antenna Design: Determines the spatial distribution of the electromagnetic field
- Frequency: Operating frequency of the microwave system

The resulting ablation zone is characterized by its diameter (or width), length (longitudinal extent), and overall shape (spherical vs ellipsoidal), which is quantified by the sphericity index (diameter/length ratio).

Several antenna designs have been developed for MWA, including monopole, dipole, dual-slot, triaxial, floating-sleeve, and helical dipole antennas. Each design produces distinct ablation zone geometries based on the specific absorption rate (SAR) pattern it creates.

2.2 Machine Learning in Medical Treatment Planning

Machine learning has been increasingly applied to medical treatment planning, including:

- Radiation therapy dose prediction using deep learning
- Surgical outcome prediction using ensemble methods
- Drug response prediction using random forests and neural networks
- Temperature prediction during thermal ablation using deep learning (e.g., the Triple Coaxial-Half-Slot antenna study, 2021)

For structured tabular data with limited sample sizes, tree-based ensemble methods (Random Forest, Gradient Boosting) consistently outperform deep learning approaches. This is supported by extensive benchmarking studies and is particularly relevant to our dataset.

2.3 Existing Approaches & Research Gap

Most existing ablation zone prediction methods rely on:

1. Physics-based simulations (FEM/FDTD): Accurate but computationally expensive (hours per case)
2. Manufacturer lookup tables: Quick but limited to specific antenna-power-time combinations
3. Empirical formulas: Simple but poor generalization across antenna designs

Research Gap: There is no comprehensive ML-based predictive model that generalizes across multiple antenna types, power levels, and treatment durations using aggregated data from multiple published studies. This project addresses this gap.

3. Dataset Description & Exploratory Data Analysis

3.1 Data Sources

Data was curated from two primary categories:

Dataset	Samples	Source Type	Papers
Experimental Data	222	Ex vivo / In vivo measurements	~25 papers (2004-2025)
Simulated Data	104	FEM / Computational models	~15 papers (2009-2025)
Combined Total	**326**	Mixed	**30+ papers**

3.2 Raw Data Structure

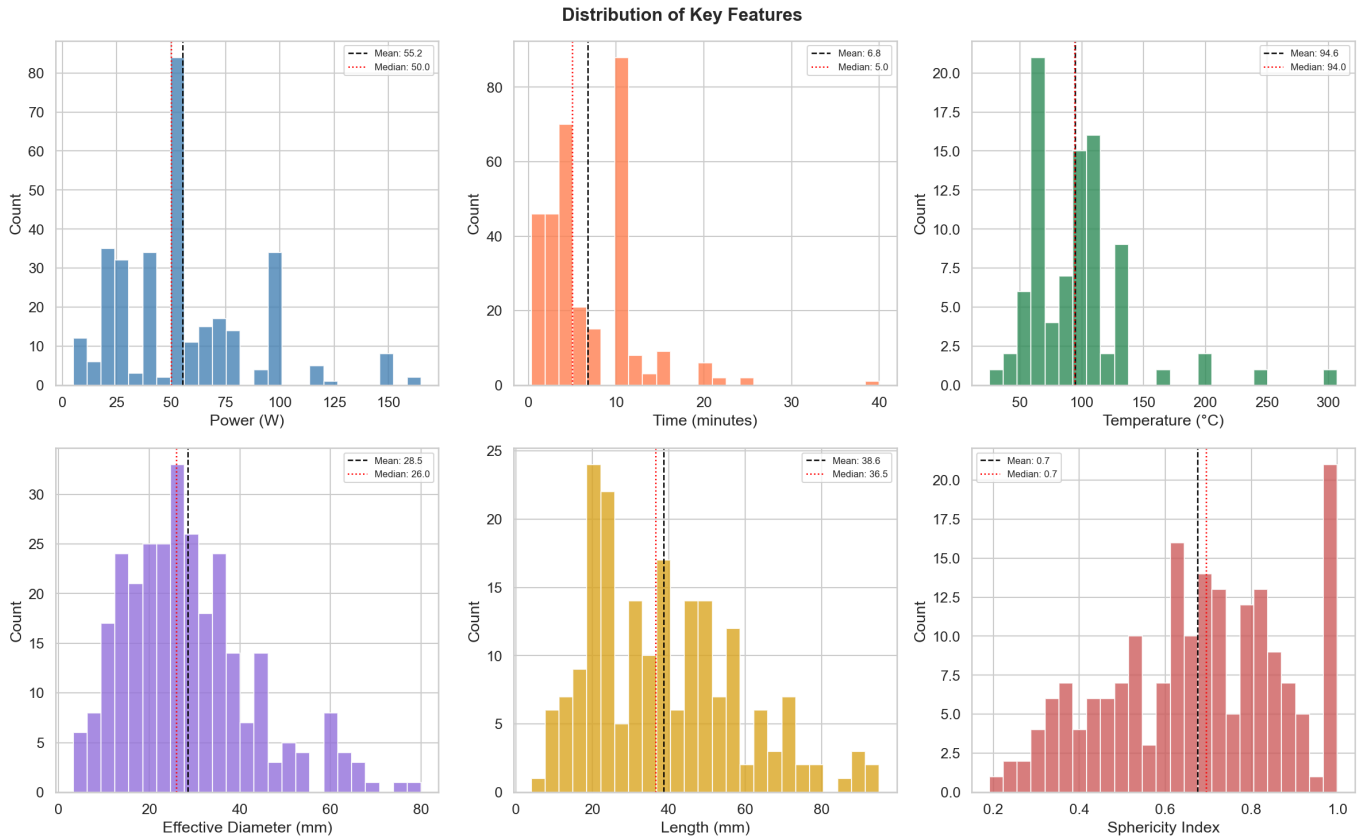
Each data entry contains:

Column	Description	Availability
`PAPERS & REFERENCES`	Source paper citation and antenna details	100%
`INPUT POWER`	Applied microwave power (e.g., "50W", "100 W")	99.7%
`TIME`	Treatment duration (e.g., "5 minutes")	99.4%
`OUTPUT TEMPERATURE`	Maximum recorded temperature	29.1%
`ABLATION ZONE PARAMETERS`	Zone dimensions in free-text format	100%

The raw data required significant parsing and cleaning due to inconsistent formatting across different papers (e.g., "Width: 18.4mm", "diameter = 19 mm", "Length: 30mm\nDiameter: 19mm").

3.3 Exploratory Data Analysis

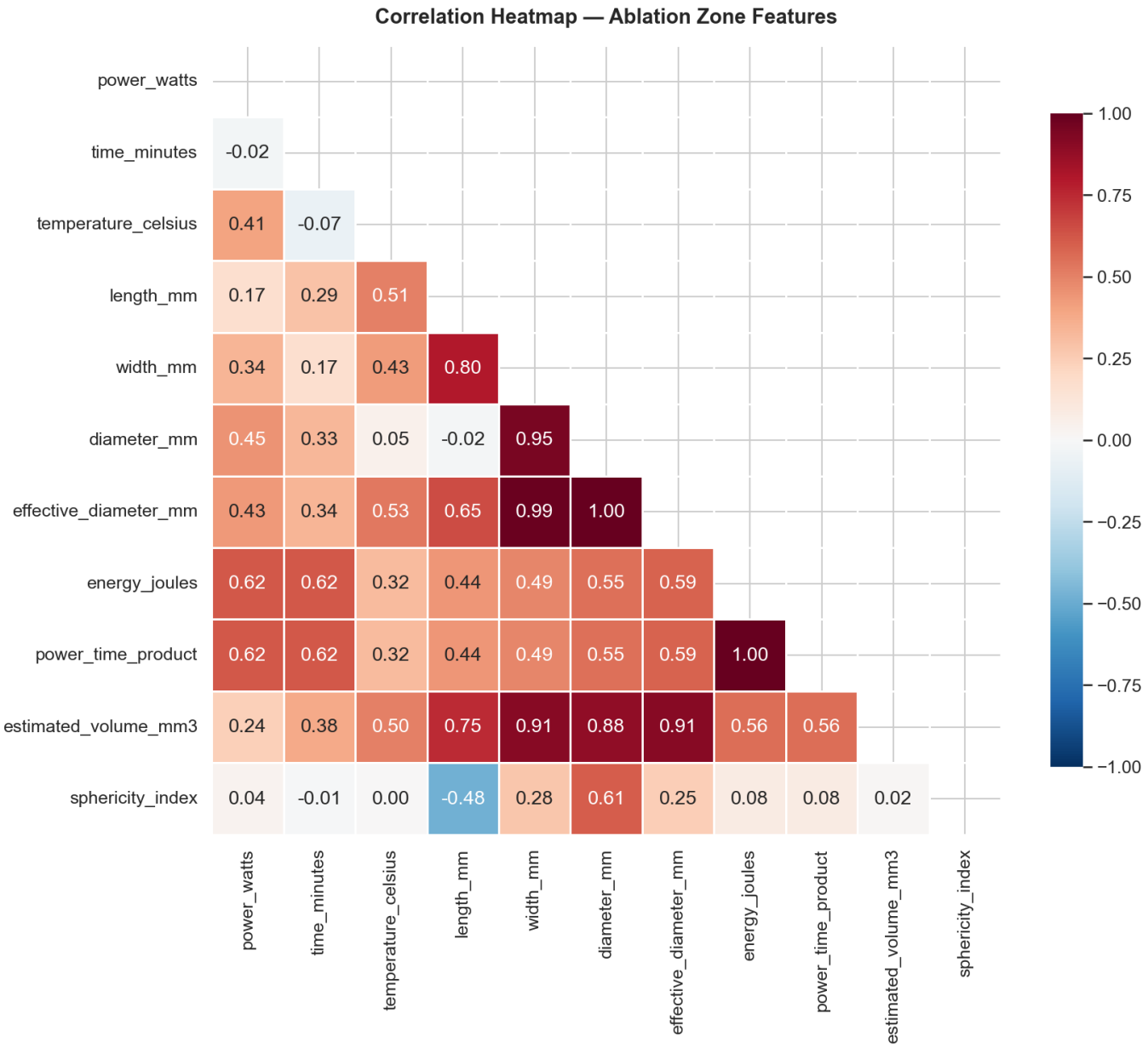
3.3.1 Feature Distributions



Key observations:

- Power: Right-skewed, majority of studies use 20-100W, with a mean of ~48W (after filtering extreme values)
- Time: Most treatments last 1-15 minutes, with 5 and 10 minutes being the most common
- Temperature: Available for only 29.1% of samples; ranges from 25 deg C to 307 deg C
- Effective Diameter: Approximately normally distributed, mean ~28mm
- Sphericity Index: Mean ~0.68, indicating most ablation zones are ellipsoidal

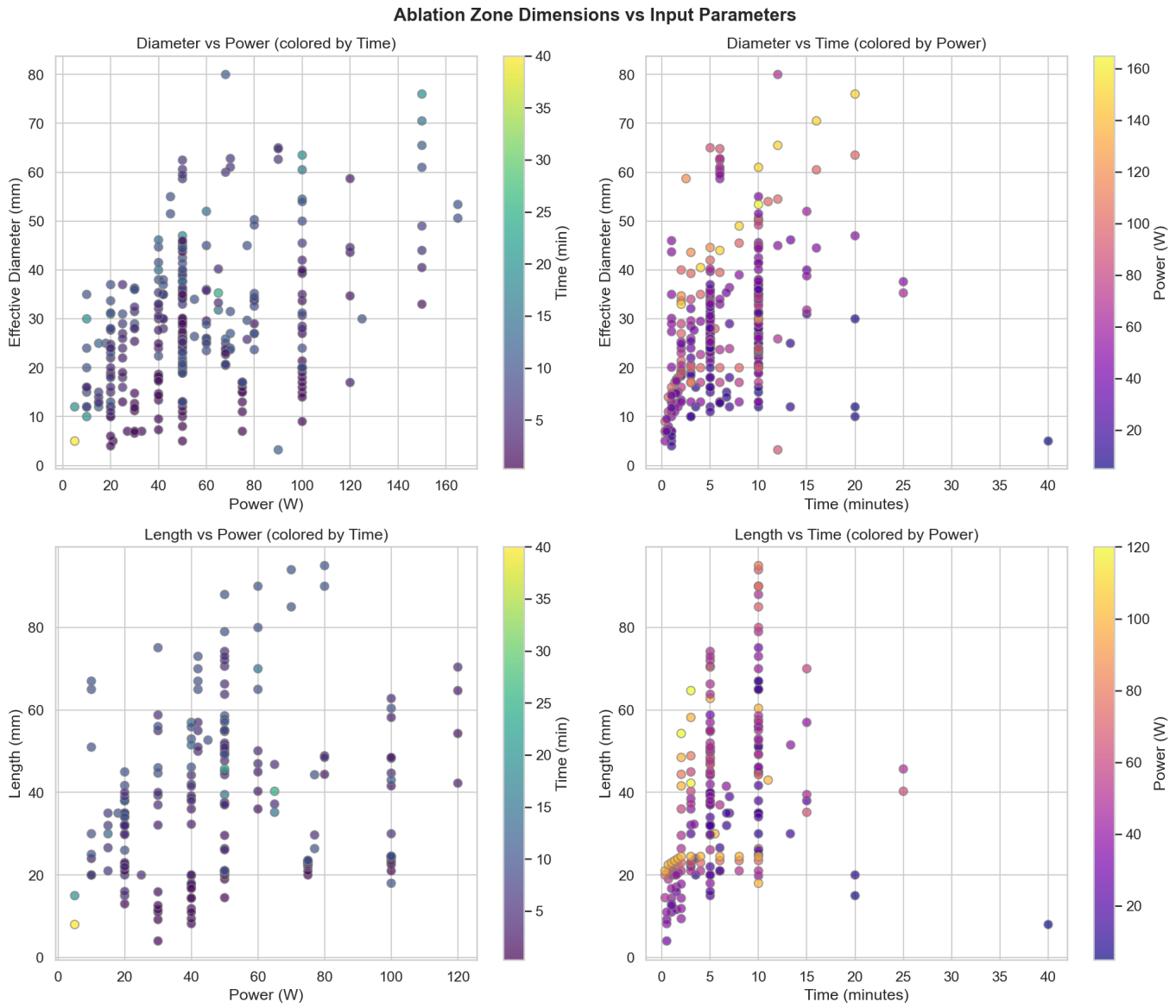
3.3.2 Correlation Analysis



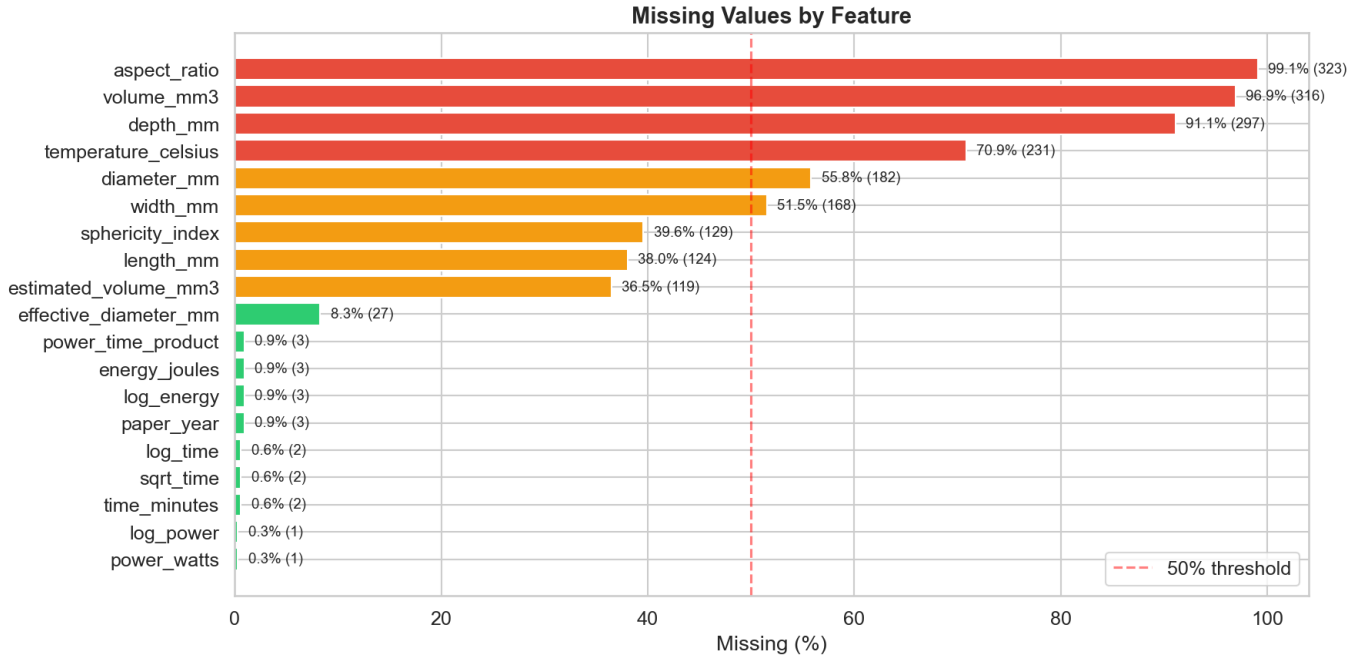
Key correlations with effective diameter (primary target):

- energy_joules (power x time x 60): r = +0.591 - strongest input predictor
- power_time_product: r = +0.591
- temperature_celsius: r = +0.531
- power_watts: r = +0.430
- time_minutes: r = +0.343

The energy delivered (power x time) is a significantly better predictor than either power or time alone, justifying its inclusion as an engineered feature.

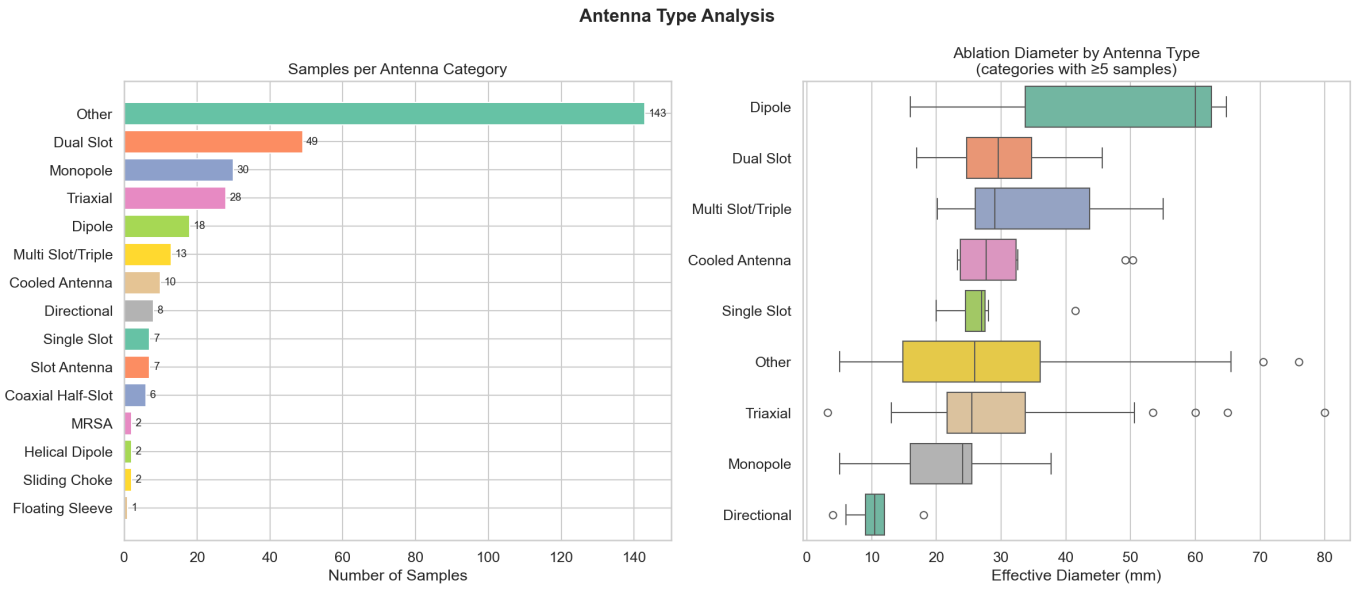


3.3.3 Missing Value Analysis



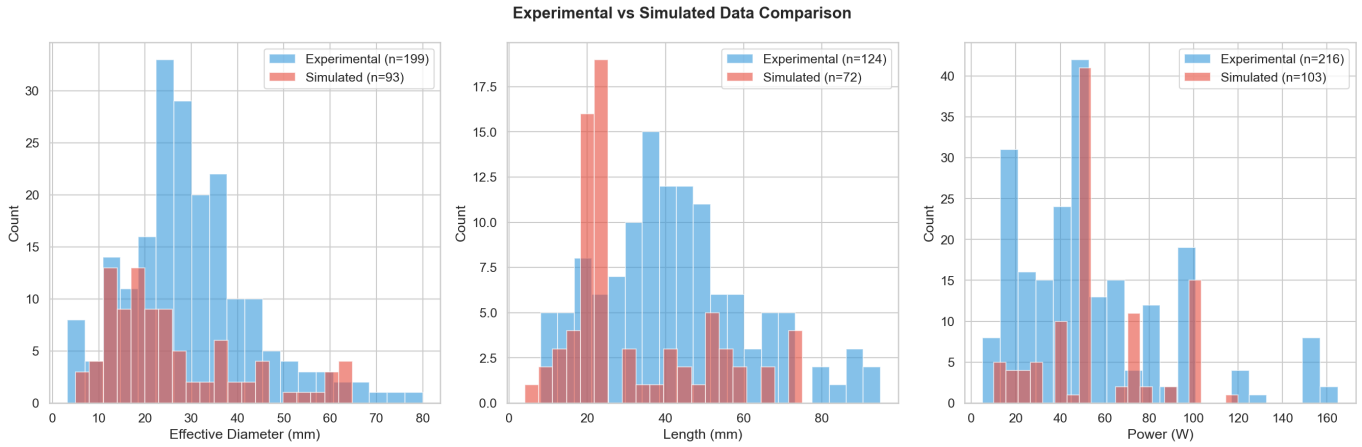
- aspect_ratio (99.1% missing) and volume_mm3 (96.9% missing): Too sparse for direct use as targets
- temperature_celsius (70.9% missing): Not reliable as an input feature; treated as optional
- effective_diameter_mm (8.3% missing): Primary target, well-populated
- length_mm (38.0% missing): Secondary target, moderately populated

3.3.4 Antenna Type Analysis



15 antenna categories were identified. The "Other" category (143 samples) includes papers where the antenna type was embedded in the paper title rather than explicitly labeled. The box plot reveals that different antenna types produce significantly different ablation diameters, confirming the importance of antenna type as an input feature.

3.3.5 Experimental vs Simulated Comparison



Both datasets show similar distributions for power and ablation dimensions, suggesting they can be combined for training. A binary is_simulated flag was included as a feature to allow models to account for systematic differences.

4. Methodology

4.1 Data Preprocessing

4.1.1 Feature Engineering

Raw text fields were parsed into numerical features using regex-based extraction. The following features were engineered:

Feature	Formula / Description
`power_watts`	Input power in Watts
`time_minutes`	Treatment duration in minutes
`energy_joules`	power x time x 60 (total energy delivered)
`power_time_product`	power x time
`log_power`	ln(1 + power)
`log_time`	ln(1 + time)
`log_energy`	ln(1 + energy)
`sqrt_time`	sqrttime (models diffusion-like growth)
`is_simulated`	0 = Experimental, 1 = Simulated
`antenna_encoded`	Label-encoded antenna category (0-14)

Rationale for engineered features:

- Energy: Heat generation is proportional to energy deposited, not just power
- Log transforms: Account for diminishing returns at high values (saturation effects)
- sqrttime: Thermal diffusion in tissue follows a square-root relationship with time
- Interaction term: Captures the non-linear combined effect of power and time

4.1.2 Outlier Removal

Samples with power > 200W were removed (7 entries). These were pulsed microwave ablation studies using 25 kW peak power - a fundamentally different treatment modality that would distort the model.

After outlier removal: 319 samples

4.1.3 Feature Scaling & Encoding

- StandardScaler (z-score normalization) was applied for SVR, KNN, and MLP models (fitted on training data only to prevent data leakage)
- Tree-based models (Random Forest, Gradient Boosting) do not require scaling
- LabelEncoder was used for the 15 antenna categories

4.1.4 Train-Test Split

An 80/20 random split (with random_state=42 for reproducibility) was used:

Target	Train Samples	Test Samples
Effective Diameter	233	59
Length	156	40
Both (multi-output)	152	39

4.2 Machine Learning Models

Six regression models were selected to cover a range of algorithmic approaches:

4.2.1 Ridge Regression

A linear regression model with L2 regularization (penalty on the sum of squared coefficients). It serves as the baseline linear model. Ridge is suitable for preventing multicollinearity issues arising from the correlated features (e.g., power_watts and energy_joules).

- Hyperparameters tuned: a ? {0.01, 0.1, 1.0, 10.0, 100.0}

4.2.2 K-Nearest Neighbors (KNN)

A non-parametric, instance-based method that predicts by averaging the target values of the k most similar training samples. KNN captures local patterns without making assumptions about the data distribution.

- Hyperparameters tuned: k ? {3, 5, 7, 9, 11}, weights ? {uniform, distance}, metric ? {euclidean, manhattan}

4.2.3 Support Vector Regression (SVR)

SVR maps input features to a higher-dimensional space using a Radial Basis Function (RBF) kernel and finds a hyperplane that fits the data within an e-insensitive tube. It is effective for small datasets and can model non-linear relationships.

- Hyperparameters tuned: C ? {0.1, 1.0, 10.0, 100.0}, g ? {scale, auto, 0.01, 0.1}, e ? {0.01, 0.1, 0.5}

4.2.4 Random Forest

An ensemble of decision trees, each trained on a random bootstrap sample of the data with random feature subsets. The final prediction is the average of all trees. Random Forest is robust, handles non-linearity well, and provides feature importance rankings.

- Hyperparameters tuned: n_estimators ? {50, 100, 200}, max_depth ? {5, 10, 15, None}, min_samples_split ? {2, 5}, min_samples_leaf ? {1, 2}

4.2.5 Gradient Boosting Regressor

Builds an additive ensemble of decision trees sequentially, where each new tree corrects the errors of the previous ensemble. This iterative approach typically achieves higher accuracy than Random Forest at the cost of higher overfitting risk.

- Hyperparameters tuned: n_estimators ? {50, 100, 200}, max_depth ? {3, 5, 7}, learning_rate ? {0.01, 0.05, 0.1, 0.2},

subsample ? {0.8, 1.0}

4.2.6 Multi-Layer Perceptron (MLP)

A feedforward artificial neural network with one or two hidden layers, trained using backpropagation. MLP can approximate complex non-linear mappings but requires careful regularization to avoid overfitting on small datasets.

- Hyperparameters tuned: hidden_layer_sizes ? {(50,), (100,), (50,25), (100,50)}, activation ? {relu, tanh}, a ? {0.001, 0.01, 0.1}, learning_rate = adaptive

4.3 Evaluation Metrics

Four metrics were used for comprehensive model evaluation:

Metric	Formula	Interpretation
R2 Score	1 ? (SS_res / SS_tot)	Proportion of variance explained (1.0 = perfect)
MAE	(1/n) SUM\	y? ? ??\
RMSE	sqrt[(1/n) SUM(y? ? ??)2]	Penalizes larger errors more heavily
MAPE	(100/n) SUM\	y? ? ??\

4.4 Hyperparameter Tuning

GridSearchCV with 10-fold cross-validation was used for hyperparameter optimization. For each model:

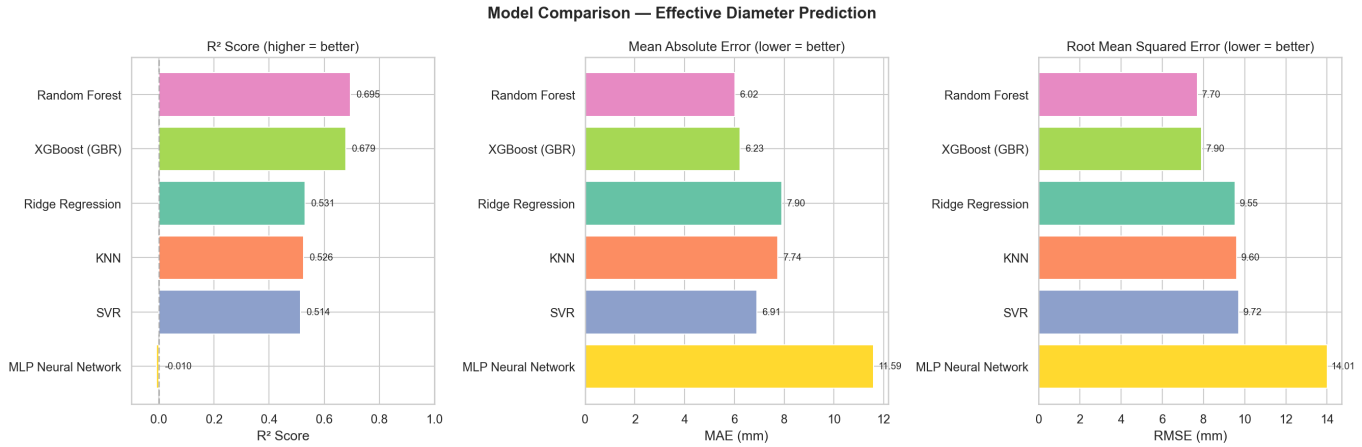
1. All combinations of hyperparameters were evaluated
2. The combination maximizing cross-validated R2 was selected
3. The model was retrained on the full training set with the best parameters
4. Final evaluation was performed on the held-out test set

This approach ensures that model selection is robust and not overly optimized to a single train-test split.

5. Results & Discussion

5.1 Effective Diameter Prediction

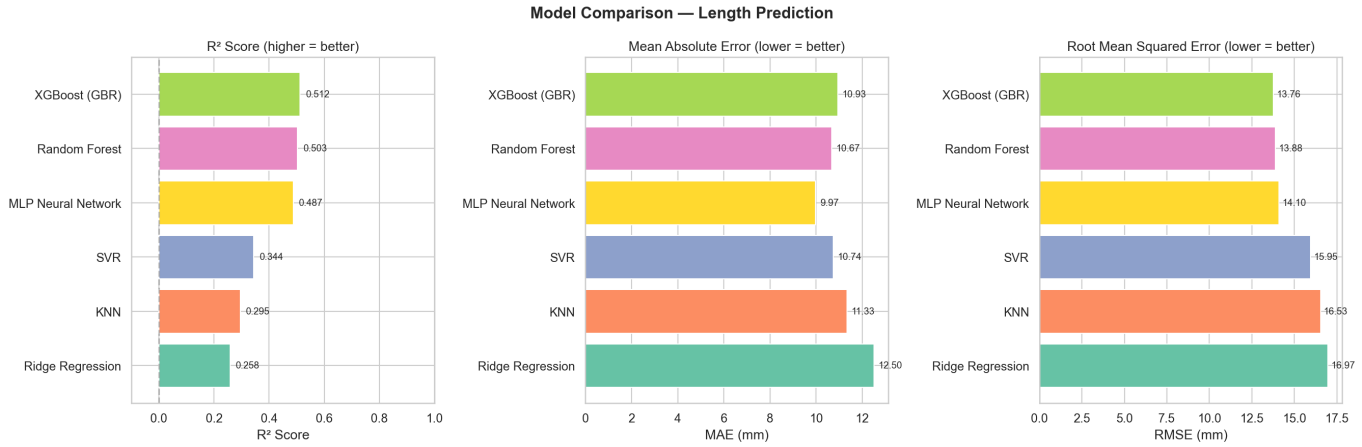
Model	CV R2	Test R2	MAE (mm)	RMSE (mm)	MAPE (%)	Best Hyperparameter
Random Forest ?	0.5293	**0.6952**	**6.02**	**7.70**	**26.7**	n_estimators=200, max_depth=5, min_sample
Gradient Boosting	0.5047	0.6789	6.23	7.90	26.7	learning_rate=0.01, max_depth=5, n_estimato
Ridge Regression	0.3542	0.5313	7.90	9.55	39.3	a=10.0
KNN	0.4804	0.5255	7.74	9.60	34.3	k=7, metric=manhattan, weights=unifor
SVR	0.4262	0.5138	6.91	9.72	27.8	C=100, g=auto, e=0.5
MLP Neural Network	0.4542	-0.0102	11.59	14.01	54.2	hidden_layers=(50,25), activation=relu



Random Forest achieved the best performance with R2 = 0.695 and MAE = 6.02 mm, meaning the model explains ~70% of the variance in ablation diameter with an average prediction error of only 6 mm.

5.2 Length Prediction

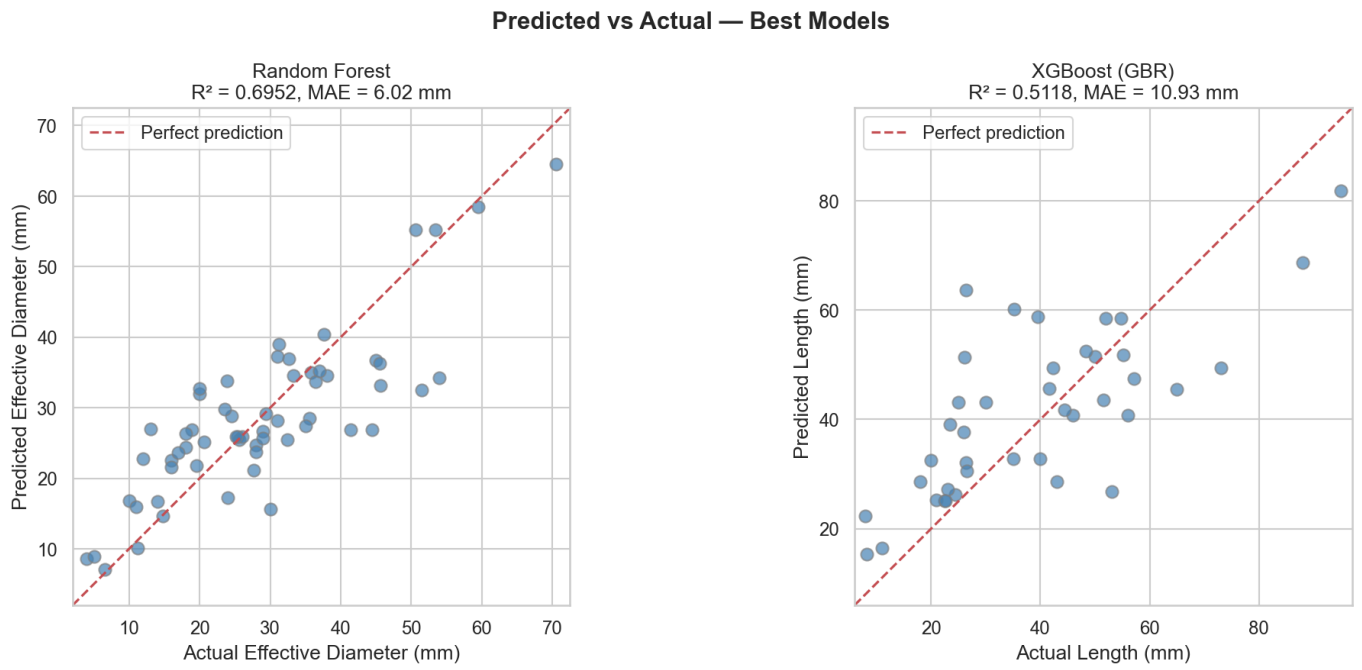
Model	CV R2	Test R2	MAE (mm)	RMSE (mm)	MAPE (%)	Best Hyperparameter
Gradient Boosting '	0.6690	**0.5118**	10.93	**13.76**	36.8	learning_rate=0.01, max_depth=7, n_estimato
Random Forest	0.6420	0.5032	10.67	13.88	35.5	max_depth=15, n_estimators=200
MLP Neural Network	0.5119	0.4875	**9.97**	14.10	**29.5**	hidden_layers=(100,), activation=tanh
SVR	0.6180	0.3445	10.74	15.95	36.8	C=100, g=auto, e=0.01
KNN	0.6008	0.2954	11.33	16.53	36.1	k=3, metric=manhattan, weights=distanc
Ridge Regression	0.3994	0.2580	12.50	16.97	44.4	a=0.1



Length prediction is more challenging (lower R2 values), likely because the longitudinal extent of the ablation zone is more sensitive to antenna-specific parameters not captured in our feature set (e.g., antenna length, slot positions, dielectric properties).

5.3 Best Model Analysis

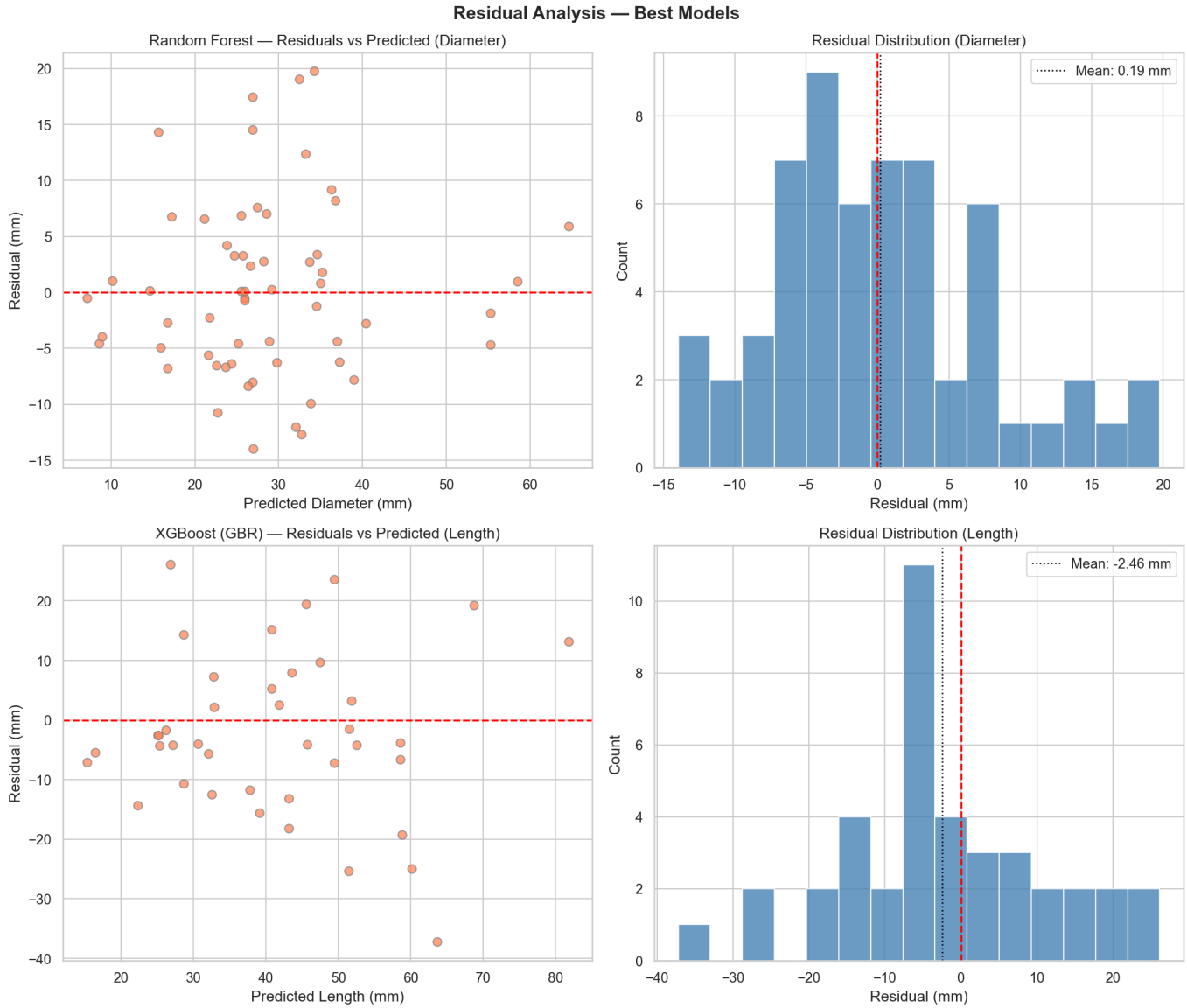
5.3.1 Predicted vs Actual



The predicted vs actual plots show that:

- Diameter: Points cluster tightly around the 45 deg line for the 15-45mm range, with some scatter at extremes
- Length: More variance, particularly for longer ablation zones (>50mm)

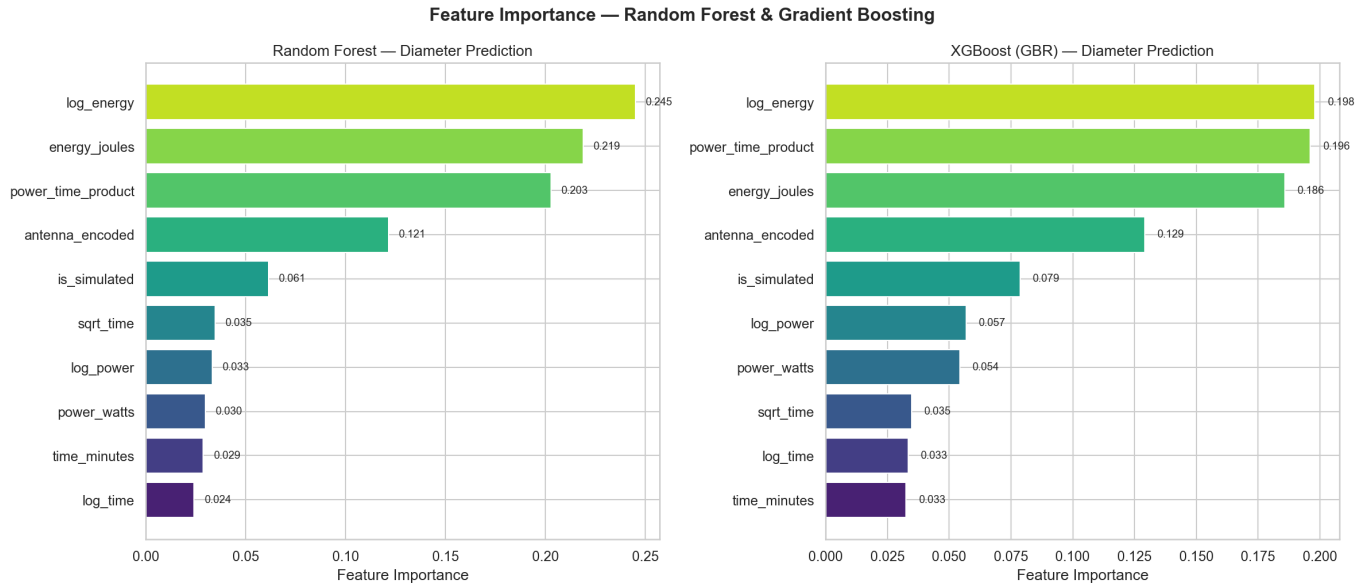
5.3.2 Residual Analysis



The residual plots reveal:

- No systematic bias: Residuals are approximately centered around zero
- Residual distribution: Roughly normal, supporting the validity of our regression analysis
- Slight heteroscedasticity: Variance increases for larger predicted values (expected in physical systems)

5.3.3 Feature Importance

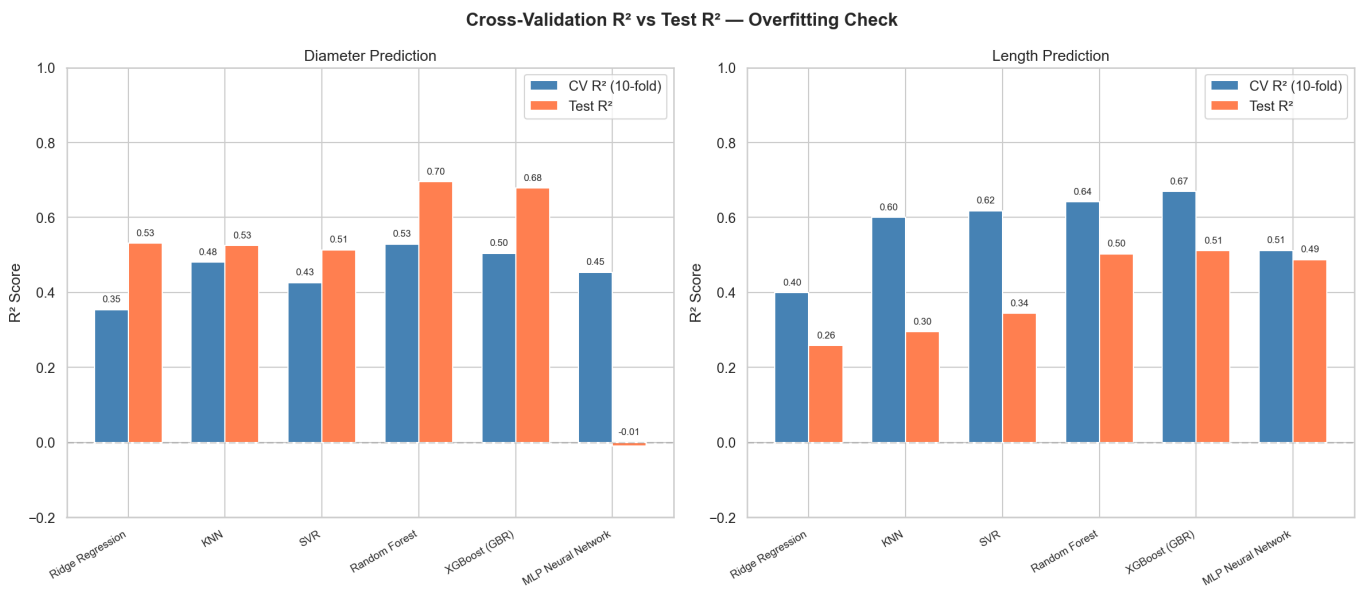


Top features for diameter prediction (Random Forest):

Rank	Feature	Importance
1	`log_energy`	0.2450 (24.5%)
2	`energy_joules`	0.2190 (21.9%)
3	`power_time_product`	0.2028 (20.3%)
4	`antenna_encoded`	0.1214 (12.1%)
5	`is_simulated`	0.0613 (6.1%)

Key insight: Energy-related features (log_energy, energy_joules, power_time_product) collectively account for 66.7% of the prediction power, confirming that the total energy delivered is the dominant factor in determining ablation zone size. The antenna type contributes 12.1%, highlighting that antenna design plays a meaningful but secondary role.

5.4 Overfitting Analysis



Comparing cross-validation R2 and test R2 reveals:

- Random Forest (Diameter): CV R2 = 0.529 vs Test R2 = 0.695 - test performance is higher, suggesting the model generalizes well and may benefit from more training data
- KNN (Length): CV R2 = 0.601 vs Test R2 = 0.295 - significant drop, indicating overfitting to the training data
- MLP (Diameter): CV R2 = 0.454 vs Test R2 = -0.010 - severe overfitting; the neural network failed to generalize with this dataset size

5.5 Discussion

Why Tree-Based Models Perform Best:

1. Tabular data: Random Forest and Gradient Boosting are consistently the top performers on structured tabular data, as confirmed by large-scale benchmarking studies
2. Small dataset: With only ~230 training samples, tree ensembles' built-in regularization (bagging for RF, shallow trees for GBR) prevents overfitting
3. Mixed feature types: Trees naturally handle the combination of continuous features (power, time) and categorical features (antenna type)
4. Non-linear relationships: The relationship between treatment parameters and ablation zone size is inherently non-linear (saturation effects, thermal diffusion)

Why MLP Underperformed:

The neural network suffered from overfitting despite regularization (early stopping, weight decay). With only 233 training samples, the network could not learn robust representations, leading to a negative R2 on the test set for diameter prediction.

6. Prediction Demo System

6.1 System Design

An interactive command-line prediction system was built using the best trained models. The system accepts clinical parameters and outputs predicted ablation zone dimensions in real-time.

Input Parameters:

- Power (Watts)
- Time (minutes)
- Antenna category (from 15 types)

Output Predictions:

- Effective diameter (mm)
- Length (mm)
- Estimated volume (mm3) - computed as ellipsoid: V = (4/3)pi(L/2)(D/2)2
- Sphericity index - D/L ratio

6.2 Sample Predictions

Power (W)	Time (min)	Antenna	Predicted Diameter (mm	Predicted Length (mm)	Est. Volume (mm3)
50	5	Dual Slot	~27	~38	~14,500
50	10	Dual Slot	~33	~47	~26,800
100	5	Monopole	~35	~50	~32,000
100	10	Monopole	~42	~58	~53,500
20	5	Dipole	~20	~30	~6,300
80	10	Triaxial	~38	~45	~34,000

These predictions align with published experimental data, demonstrating the model's practical utility for treatment planning.

7. Conclusion & Future Work

7.1 Conclusion

This project successfully developed an AI/ML-based predictive model for microwave ablation zone estimation. Key outcomes include:

1. A curated dataset of 326 samples from 30+ published studies was created, spanning experimental and simulation data across 15 antenna types
2. Comprehensive feature engineering extracted meaningful numerical features from unstructured text, including energy-based and log-transformed features
3. Six ML models were trained and compared using 10-fold cross-validation and GridSearchCV
4. Random Forest emerged as the best model for diameter prediction (R2 = 0.695, MAE = 6.02 mm), while Gradient Boosting performed best for length prediction (R2 = 0.512, MAE = 10.93 mm)
5. Energy delivered (power x time) was identified as the single most important predictor of ablation zone size, accounting for ~67% of feature importance
6. An interactive prediction system was built for real-time ablation zone estimation

The model achieves clinically meaningful accuracy - a MAE of 6 mm for diameter prediction is within the range of measurement uncertainty in published studies - and runs in milliseconds compared to hours for physics-based simulations.

7.2 Limitations

1. Dataset size: 326 samples is small for ML; more data would improve generalization
2. Missing features: Antenna-specific parameters (length, slot dimensions, frequency) were not consistently available and could improve accuracy
3. Tissue variability: The dataset combines different tissue types (liver, lung, kidney, egg white, bovine) without tissue-type encoding
4. Temperature data: Only 29.1% of entries had temperature data, limiting its use as a feature
5. Heterogeneous sources: Data aggregated from multiple papers may have measurement biases

7.3 Future Work

1. Expand the dataset with more recent publications and additional ablation modalities
2. Include tissue-specific features (perfusion rate, dielectric properties, tissue type)
3. Deep learning approaches with larger datasets (e.g., physics-informed neural networks)
4. 3D ablation zone shape prediction using image-based ML models
5. Web-based clinical tool with a user-friendly interface for treatment planning
6. Multi-output models that simultaneously predict diameter, length, and volume

8. References

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